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From olive stones waste to valuable resource: Exploring various techniques for cellulose extraction

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ABSTRACT

Cellulose was extracted from olive stones using various extraction techniques, including microwave-assisted extraction (MAE), conventional heating extraction (CHE), hot water extraction (HWE), and ultrasound-assisted extraction (UAE). These methods consisted in sequential acid treatment with hydrochloric acid (HCl), alkaline treatment with sodium hydroxide (NaOH), and bleaching with hydrogen peroxide (H_2O_2). Among them, MAE and CHE yielded significantly higher extraction yields, while reducing reaction time and solvent consumption as compared to HWE and UAE. NMR and FT-IR analyses confirmed the chemical structure of extracted cellulose and revealed the presence of residual lignin. Furthermore, celluloses obtained through MAE and CHE present higher crystallinity index and enhanced thermal stability as compared to those of HWE and UAE. These findings highlight MAE and CHE as promising techniques for cellulose extraction from olive stones in the context of circular economy, thus paving the way scaling and potential applications in the health and pharmaceutical sectors.

1. Introduction

The olive tree, found mainly in the regions surrounding the Mediterranean basin, is well known for the production of olive oil and table olives. According to the International Olive Council, more than three million tons of olive oil are produced worldwide in the 2024/2025 campaign, with Spain the first producer, followed by Türkiye, Tunisia, Greece and Italy. The health benefits of olive oil have been largely documented [1-3], particularly its richness in antioxidants (vitamin E, polyphenols, …) which contribute to eliminate free radicals involved in some chronic diseases and ageing, and thus to extend life expectancy [4]. Nevertheless, olive oil production generates huge amounts of by-products and wastes, such as aqueous olive mill wastewater, cake, pomace and stones.

Olive stones are mainly composed of lignocellulosic materials, with 31.9 % of cellulose, 26.5 % of lignin and 21.9 % of hemicellulose [5], with remaining 20 % of fats and proteins. Cellulose is a polysaccharide composed of repeated unbranched β (1−4) D-glucose units. It is the most abundant natural polymer in the world, located mainly in the cell wall of plants [6]. Its high crystallinity and strong hydrogen bonding between linear chains render cellulose insoluble in water. As a biopolymer, cellulose present outstanding biocompatibility and biodegradability, making it a material of choice for applications in the food packaging, pharmaceutical and biomedical industries. Various extraction techniques are available for isolating polysaccharides directly from plants [7-9], and fruits [10-13], including steam explosion [7], acid catalyzed hydrolysis [8], autoclave-assisted alkali treatment [10], hot water extraction (HWE), ultrasound-assisted extraction (UAE), microwave-assisted extraction (MAE) and conventional heating extraction (CHE).

HWE and UAE alternate acid and base treatments, followed by bleaching to recover the final product [6,14,15]. These two methods differ in the pre-treatment process: by increasing the temperature to the boiling point for HWE and using ultrasound for UAE [16,17], the cell wall is broken, allowing progressive release of polysaccharides [18-20]. Garnett et al. used the hot water extraction method to extract cellulose from fruit and vegetable wastes collected in a local restaurant. The extracted cellulose exhibits a chemical structure similar to commercial

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products, highlighting the effectiveness of this extraction technique in cellulose isolation [6]. HWE and UAE methods are simple, easily scalable and eco-friendly [20]. Conversely, MAE and CHE methods require shorter heating time and smaller volume of solvent due to harsher conditions to which the raw material is subjected (using autoclave to heat the solution in a closed vessel) [21]. These techniques enable more targeted extraction of the desired polysaccharide [22]. Vinhas et al. used a synthesis microwave oven to extract cellulose from waste eucalyptus and pine wood. The extraction process consists in three steps: 1) alkaline treatment using NaOH; 2) bleaching I using H_2O_2; and 3) bleaching II using H_2SO_4. In this process, alkaline treatment (step 1) swells the lignocellulosic fibers, leading to structural degradation and elimination of hemicellulose and part of lignin [23]. As lignin is known to have good resistance to chemical and thermal attack, steps 2 and 3 are necessary to complete the removal of lignin. Thanks to the harsh extraction conditions (microwave irradiation, strong alkaline and acidic solutions, high temperature up to 120 °C in a total of 3 h), cellulose was obtained at high yields with removal of all hemicellulose and part of lignin [22]. On the other hand, few data are available in literature on the use of conventional heating using autoclaves for cellulose extraction probably due to poor control of temperature and power parameters in contrast to the microwave method. The present work focused on the development of a protocol for efficient cellulose extraction using resistance heating (CHE), in comparison with more widely used methods such as HWE, UAE and MAE.

The valorization of olive stones from industrial production is attracting growing attention, but it still remains under-exploited. Waste products from olive oil production such as pomace (skin, pit, pulp and kernel) have been the subject of several studies [24-26]. Katsina et al. extracted phenolic compounds from pomace for use in ophthalmic applications [27]. Aliakbarian et al. prepared maltodextrin based nanoparticles encapsulating polyphenols extracted from olive pomace using supercritical assisted atomization (SAA) for uses in the food, pharmaceutical and cosmetics industries [28]. In the energy field, Torres-Bujalane et al. reported the synthesis of phosphorus-modified porous carbons from olive stones and their use as support of Pd particles for the oxidative condensation of furfural with ethanol [29]. Others studies focused on the formation of biochar from olive stones for environmental applications [30] or for CO_2 methanation [31]. There has been a great deal of research on strategies of adding value to this product in the energy, environmental and agri-food sectors [32]. Nevertheless, few studies have been reported in literature on the biomedical applications of cellulose extracted from olive stones, so far.

In this work, HWE, UAE, MAE and CHE methods were comparatively used to isolate cellulose from olive stones so as to figure out if emerging techniques such as MAE and CHE bring advantages compared to most widely used methods (HWE and UAE). The extracted celluloses were characterized using various analytical techniques to determine their chemical structure and physico-chemical properties. The aim of this work was to add value to the abundant by-products of olive industry, specifically olive stones, in a circular economy approach for further scaling and potential applications in the health and pharmaceutical sectors.

2. Materials and methods

2.1. Materials

Olive stone samples from different olive varieties were obtained from BOUKHRIS Oil Mill (Sfax, Tunisia), then dried and milled into olive stone flour (OSF) prior to analysis. Three types of samples were used: Chemlali (Chem_OSF), Koroneiki (Kor_OSF) and a mixture of several varieties (Mix_OSF). The latter consisted of a blend of Chemlali, Chetoui, Koroneiki, and Arbequina varieties, which were co-processed during the same extraction campaign to reflect typical industrial conditions. Total dietary fiber (TDF) analysis kit, hydrochloric acid (HCl), sodium hydroxide (NaOH), hydrogen peroxide 35 % (H_2O_2), sulfuric acid (H_2SO_4) and 1-ethyl-3-methylimidazolium chloride were provided by Sigma Aldrich. All chemicals were of analytic grade and used without further purification.

2.2. Cellulose extraction

2.2.1. Hot water extraction (HWE)

Hot water extraction was carried out based on the method reported by Garnett et al. [6]. OSF (1 g) was suspended in distilled water (6 mL), boiled for 10 min, and filtered. The residue was then treated with 1 M HCl (6 mL), stirred at 85°C for 30 min, and filtered again. This acidic treatment was repeated using 0.5 M HCl. Then, the obtained residue was subjected to three successive alkaline treatments with 6 mL of 1 M NaOH, each stirred at 85°C for 30 min and filtrated. Bleaching was then carried out twice using 3 % hydrogen peroxide, followed by 1 h of stirring at 70°C and filtration. The final product was washed with distilled water until a neutral pH was reached and then lyophilized.

2.2.2. Ultrasound assisted extraction (UAE)

OSF (1 g) was suspended in 25 mL of distilled water and subjected to ultrasound treatment at 70°C for 20 min at a frequency of 35 kHz. The resulting mixture was then processed following the same protocol described in Section 2.2.1, including successive acidic and alkaline treatments, followed by bleaching steps.

2.2.3. Microwave assisted extraction (MAE)

MAE was carried out according to the method of Vinhas et al. [22]. A suspension of 3 g of OSF in 30 mL of 2 % (w/v) NaOH was transferred to a Teflon™ pressure vessel and heated for 1 h at 120°C (P = 600 W, 44 °C/min) using a Microsynth Milestone microwave oven equipped with a HPR 1000/10 rotor. The final suspension was filtered and the solid residue was washed with water and ethanol successively until a clear filtrate with neutral pH was obtained. The solid was dried in vacuum chamber at 65°C overnight.

A first bleaching step of the as-obtained sample was carried out in 30 mL of a 30 % (w/w) hydrogen peroxide solution. The mixture was heated in the same microwave oven for 1 h at 50°C (P = 250 W, 44 °C/min). The suspension was filtered, and washed successively with water and ethanol until a pH was reached, followed by a vacuum drying at 65°C overnight.

A second bleaching step was carried out by transferring the dried sample in 30 mL solution containing 1 M sulfuric acid and 10 % (v/v) 1-ethyl-3-methylimidazolium chloride followed by heating in the microwave oven for 1 h at 95°C (P = 400 W, 44 °C/min). Finally, the sample was filtered and dried under the same conditions as described above.

2.2.4. Conventional heating extraction (CHE)

CHE was performed using similar alkaline treatment and bleaching steps as those used in MAE. The OSF containing suspensions were placed in a hydrothermal synthesis reactor with a PTFE-lined tank, known as autoclave, and the reactor was placed in an oven. Each treatment step was maintained at the preset temperature for 2 h before proceeding to the next step.

2.2.5. Extraction yield

The extraction yield was determined based on Eq. (1):

$$Yield(\%) = \frac{weight\ of\ dried\ extracted\ cellulose}{weight\ of\ OSF\ sample} \times 100 \quad (1)$$

This calculation allowed for the comparison of the efficiency of different extraction techniques in isolating cellulose from OSF.

2.3. Characterization of olive stones flour and extracted cellulose

2.3.1. Proximate composition analysis

The proximate composition of OSF was determined following the AOAC (Association of Official Analytical Chemists) methods. The moisture content was determined gravimetrically by drying the sample in an oven at 105°C for 24 h until constant weight (AOAC 930.15). Total mineral content (ash) was determined gravimetrically by incinerating the dried sample in a muffle furnace at 550°C until a constant weight was obtained (AOAC 942.05). Protein content (total nitrogen, N x 6.25) was estimated using the Kjeldahl method (AOAC 978.04) for the analysis of nitrogen. The fat content was estimated using Soxhlet extraction with n-hexane as solvent (AOAC 2003.05). The carbohydrates content was determined by the phenol-sulfuric acid colorimetric method [33], using D-glucose as the standard. The total dietary fiber (TDF) was determined following the official method of the AOAC (16th edition) by using a Total Dietary Fiber Assay Kit (TDF100A Sigma-Aldrich). Elemental composition (C, H, N, S) was measured using a Leco/CNHS Elemental Analyzer.

2.3.2. pH and electrical conductivity measurements

The pH and electrical conductivity of OSF were determined using a pH meter model Istek-NeoMet and a conductometer model CON-SORT C 831, respectively.

2.3.3. Chemical and structural characterization

The OSF and cellulose composition was analyzed by attenuated total reflectance Fourier transform infrared (ATR-FTIR) spectroscopy (Thermo Nicolet Nexus). The resolution was 4 cm^{-1}, and 64 scans were made between 4000 and 650 cm^{-1} for each analysis.

^{13}C solid-state nuclear magnetic resonance spectroscopy (NMR) was carried out at the Analysis and Characterization Platform (UAR 2041, Chimie Balard, Montpellier, France), using a 300 MHz Varian VNMRS300 spectrometer ("Wide Bore" magnet at 7.05 Tesla) equipped with a Varian T3 MAS (Magic Angle Spinning) probe and 3.2 mm ZrO_2 rotors. ^{13}C spectra were acquired using the non-quantitative CPMAS technique (Cross Polarization Magic Angle Spinning) with ^{1}H decoupling at a spinning frequency of 12 kHz, with a recycle delay of 3 s, a contact time of 1 ms and a $\pi/2$ pulse of 5 μs (90°). The width of the spectral window was 50 kHz and the line broadening was 30 Hz. Adamantane was used as a secondary reference for calibration (signal at 38.5 ppm).

2.3.4. Scanning electron microscopy (SEM)

The surface morphology was examined by using Hitachi S-4800 microscope, operating at spatial resolution of 1.50 nm at 15 kV for various OSF samples, and Zeiss Sigma300 at 5 kV for the cellulose samples. Samples were platinum-sputtered before analysis.

2.3.5. Thermogravimetric analysis

The thermal stability of the OSF and cellulose samples were determined by thermal gravimetric analysis (TGA) using SDT Q600 thermal analysis device (TA Instruments, USA). TGA was used to determine the decomposition temperature. Samples of 3–4 mg were heated from room temperature to 800°C at a heating rate of 10 °C/min under air atmosphere for OSF samples and under nitrogen atmosphere for cellulose samples.

2.3.6. X-ray diffraction (XRD)

X-ray diffraction (XRD) experiments were performed using the Anton Paar SAXSpoint 5.0 laboratory beamline system. Powders were put between Kapton foils and X-ray diffractograms were measured at a source to sample distance of 50.9 cm. The Primux 100 micron Cu source was operated at 50 W to generate X-rays with a wavelength of 0.15418 nm. An Eiger2 R 1 M detector was used to collect the scattered photons. For each sample, four 10 min frames were collected and averaged before azimuthal integration. A similar data collection was performed on the empty cell, containing only Kapton foils, and the integrated signal was subtracted from that of the filled cell. All measurements were done at room temperature. The crystallinity index (CI) was then calculated using Eq. 2: [34,35]

$$CI(\%) = \left(\frac{I_{002} - I_{AM}}{I_{002}}\right) \times 100\% \quad (2)$$

With I_{002} and I_{AM} corresponding to the intensity of the most intense peak and the maximum intensity of the amorphous zone, respectively.

3. Results and discussion

3.1. Characterization of olive stones flour

3.1.1. Proximate composition

The proximate composition of olive stone flours (OSF) was analyzed to evaluate their suitability for cellulose extraction, revealing significant differences among the three samples (Mix_OSF, Chem_OSF, and Kor_-OSF) in terms of moisture content, macronutrient composition, and polyphenol levels (Table A.1).

Kor_OSF exhibited the highest moisture content (11.61 ± 0.16 %), which may impact storage stability and increase energy consumption during drying, whereas Mix_OSF (7.13 ± 0.05 %) and Chem_OSF (6.92 ± 0.06 %) showed lower moisture levels, making them more favorable for processing. The high total dietary fiber content in all samples (76–86 %) suggests a promising cellulose yield, with Mix_OSF (86 %) and Kor_OSF (84 %) being particularly rich sources. Although this study primarily aims at extracting cellulose, TDF was selected as a practical and standardized indicator for evaluating the overall structural carbohydrate content, including cellulose, hemicellulose, and lignin, which are the main components of plant cell walls. TDF analysis offers a rapid and cost-effective approach for early-stage screening of biomass sources. Furthermore, Mix_OSF displayed the highest fat content (1.51 ± 0.01 %) among the samples, but its relatively low protein content (1.27 ± 0.02 %) minimizes the need for extensive purification steps, in contrast to Kor-OSF which exhibited the highest protein level (3.0 ± 0.04 %). The lower protein content in Mix_OSF can be attributed to its multi-varietal composition, including cultivars such as Chemlali, Chetoui, and Arbequina, which are known to have relatively low protein concentrations in their pits.

Furthermore, the moderate polyphenol content (1.49 ± 0.084 %) of Mix_OSF is less likely to interfere with chemical treatments during extraction, unlike the higher polyphenol concentrations observed in Kor_OSF (2.39 ± 0.2 %) and Chem_OSF (1.83 ± 0.07 %).

Overall, Mix_OSF stands out as the most favorable raw material for cellulose extraction, offering an optimal balance of high fiber content, low protein and polyphenol levels, and favorable moisture properties, all of which contribute to efficient processing with maximum extraction yield and purity.

The elemental composition of OSF samples was analyzed to determine their carbon (C), hydrogen (H), nitrogen (N), and sulfur (S) contents, which provides insights into their purity and suitability for various applications. Table A.2 demonstrates that the elemental composition of the three flours was relatively consistent. The high carbon (~47 %) and hydrogen (~6 %) contents indicate a well-structured organic material. Furthermore, the low levels of nitrogen (0.133–0.277 %) and sulfur (0.024–0.072 %) for all samples suggest minimal contamination, making them suitable for industrial applications where low nitrogen and sulfur contents are desirable.

Overall, the chemical composition data strongly support the potential of OSF, particularly Mix_OSF, as a sustainable and efficient raw material for cellulose extraction among other high-value industrial applications.

3.1.2. pH and conductivity analysis

The pH and conductivity of the OSF samples were analyzed to assess their chemical properties for potential applications. As indicated in Table A.3, all three OSF samples exhibited an acidic pH, generally around 5.0. This pH range is favorable for many applications, including food and cosmetic formulations, where slightly acidic environments can enhance product stability and shelf life. Particularly, such acidic pH of OSF is beneficial for cellulose extraction as it reduces the need for extensive pH adjustments during processing. Additionally, a slightly acidic environment can facilitate the breakdown of hemicellulose and lignin, making cellulose separation more efficient, as stated by Liu et al. [36].

The conductivity values suggest differences in ionic content among the flour varieties. Mix_OSF exhibited the highest conductivity (633 µS/cm), suggesting a higher mineral or ionic strength that could affect its functionality in product formulations.

3.1.3. Chemical and structural characterization of olive stones flour

The chemical structure of the OSF was examined by ATR-FTIR, as shown in Fig. 1 and Table A.4. Similar patterns are obtained for the three OSF varieties and the peaks associated with the main components of the olive stones are detected (Table A.4.). The peak at 1750 cm^{-1} corresponds to carbonyl groups of hemicellulose [37], the peak at 1590 cm^{-1} to lignin aromatic rings [38], and the peak at 1230 cm^{-1} to -C-O bonds present in cellulose, hemicellulose and lignin [37,38]. The most intense peak at 1030 cm^{-1} is attributed to hydroxyl groups and C-O-C bonds of polysaccharides [39]. The large peak at 3490 cm^{-1} is characteristic of hydroxyl groups and more specifically phenolic and aliphatic compounds [37], and the peak at 2930 cm^{-1} belong to the various -CH groups.

3.1.4. Thermal stability

The thermal stability of OSF was determined from TGA measurements, as shown in Fig. 2 and Table A.5. The three OSF samples exhibit almost the same mass loss profile (Fig. 2). The onset decomposition temperature values (257–258°C) of the samples are very close (Table A.5), and the maximum loss temperatures values were also comparable (283–288°C). Additionally, the residual mass is below 1 %. These findings highlight the fact that olive varieties have little impact on their thermal stability.

3.1.5. Surface morphology

The surface morphology of the OSF was examined by scanning electron microscopy, as shown in Fig. 3. The three samples present similar irregular structures with the presence of large grains of *c.a.* 30 µm. Additionally, the SEM images reveal the presence of numerous needle-like structures of smaller dimensions dispersed throughout the samples. These finer elements may be linked to cellulose fibers or fragmented lignocellulosic structures, and could enhance certain functional properties of OSF such as increased surface area. Nitrogen sorption measurements (data not shown) confirmed the absence of micropores in the material structure. Based on SEM images (Fig. 3), it is concluded that the material predominantly features pores and cavities in the macroporous range. Moreover, increased surface area enhances solvent accessibility and diffusion efficiency, facilitating more effective chemical or thermal treatments during cellulose extraction. Hence, these microstructural features can improve mass transfer kinetics, ultimately leading to higher extraction yields, shorter processing times, and reduced consumption of chemicals. These benefits are particularly relevant for sustainable and scalable biorefinery applications.

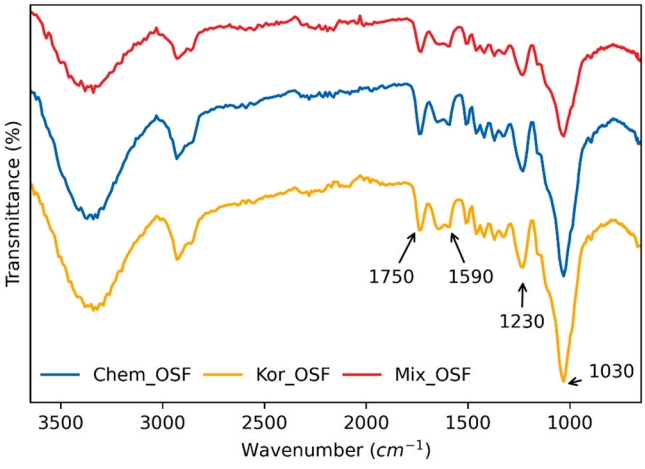


Fig. 1. FTIR spectra of the different olive stone flours (OSF) samples.

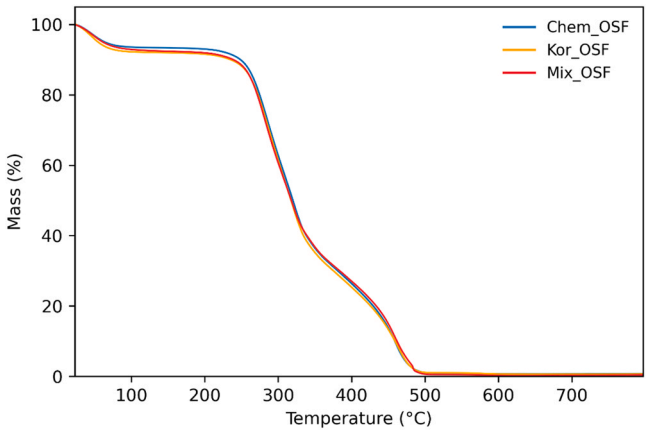


Fig. 2. TGA thermograms of olive stone flours: Chem_OSF, Kor_OSF and Mix_OSF.

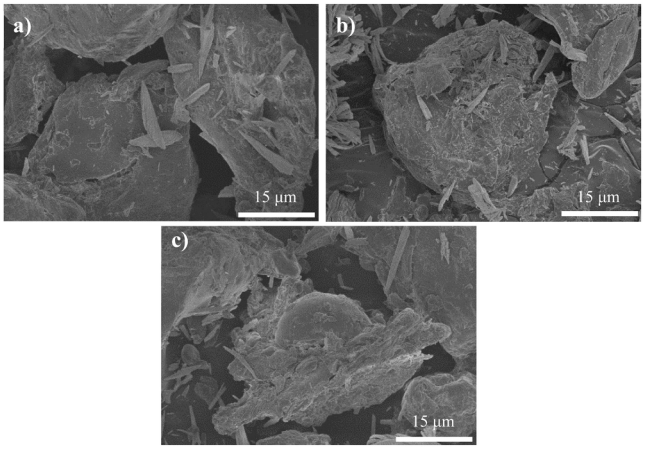


Fig. 3. SEM images of the surface of the three olive stone flours: a) Chem_OSF, b) Kor_OSF, and c) Mix_OSF.

3.2. Cellulose extraction from olive stones flour

3.2.1. Selection of raw material

The characterization of OSF prepared from different varieties demonstrated that the mixture of olive stones (Mix_OSF) is more suitable for cellulose extraction than individual varieties due to both practical and compositional advantages. Mix_OSF contains a higher total carbohydrate content, including cellulose, hemicellulose, and lignin, making it a more efficient source for cellulose recovery as evidenced by data shown in 3.1.1. Additionally, using a mixed sample eliminates the need for selective sorting, streamlining the extraction process for industrial

applications. Structural and chemical analyses, including SEM and FTIR, confirmed that Mix_OSF exhibits similar morphological and compositional properties as compared to the different varieties, ensuring consistent extraction efficiency. These findings highlight the feasibility and effectiveness of Mix_OSF as the preferred starting material for cellulose extraction in this study, enabling a more efficient, scalable, and high-yield process.

3.2.2. Extraction techniques

Various extraction techniques, *i.e.* HWE, UAE, MAE and CHE were used in order to optimize the process (Figure A.1). HWE and UAE techniques have been widely used in literature for the extraction of polysaccharides. In this work, the OSF sample was first subjected to 10 min maceration in boiling water (HWE), or 20 min ultrasonication in a water bath at 70°C (UAE). Increasing the temperature to the boiling point [40] and using ultrasound [18,19,41] resulted in the release of soluble species present inside the cell wall even before the wall structure was destroyed [42,43]. The solid residues obtained were then further treated to remove water-soluble biopolymers and residual compounds. A first acid treatment with 1 M HCl was carried out to remove pectin, followed by a second treatment with 0.5 M HCl, both for 30 min [6,44]. Alkaline treatment was then carried out 3 times using 1 M NaOH for 30 min at 85°C. This step allowed to weaken the ester bonds between hemicelluloses and cellulose, and thus to remove hemicellulose [45]. The resulting dark brown filtrate was discarded after filtration, and the remaining solid was bleached twice with hydrogen peroxide (3 %) [46,47] to remove lignin and residual impurities. This bleaching step yielded a white powder. However, despite their effectiveness, HWE and UAE techniques present some limitations, including lower extraction yields and higher solvent consumption.

Two other extraction techniques, *i.e.* microwave assisted extraction (MAE) and conventional heating extraction (CHE) using autoclaves, were also examined. Both techniques consisted in one alkaline treatment and two bleaching steps (Figure A.1). Unlike the HWE and UAE methods, the MAE and CHE techniques directly targeted the desired biopolymer. In fact, when performing the extraction in autoclaves, the autogenous pressure of solvent vapors increases the solvent's temperature beyond its boiling point at atmospheric pressure, enhancing the process efficiency. Moreover, in the case of microwave-assisted heating, the microwaves interact with both the polar solvent and cellulose chains, ensuring uniform heating throughout the entire volume of the solution. The increase in temperature and pressure within the autoclave is mainly caused by the release of heat from the polar solvent. This causes an exothermic reaction of the polar and hydroxyl groups present on the cellulose chains. Chen et al. [22,48,49]. MAE and CHE allowed to extract polysaccharides more efficiently. These techniques have the advantage of using smaller quantities of solvent and being less time consuming. The final product has a dark beige color.

The extraction yield was calculated based on Eq. 1. HWE and UAE methods led to lower extraction yields (18 % and 30 %, respectively) as compared to MAE and CHE methods (39 % and 41 %, respectively). In Section 3.1.1, it was shown that Mix_OSF sample consisted of 86 % dietary fiber, indicating that higher contents of cellulose, hemicellulose and lignin can be found compared to those obtained in the work by Rodriguez et al. with a total carbohydrate content of 80.3 % including 31.9 % cellulose [5]. This difference in yields could be related to the extraction techniques involving different extraction conditions. In fact, the HWE and UAE methods involved more steps (2 acid treatments, 3 alkaline treatments and 2 bleaching steps) than MAE and CHE methods (1 alkaline treatment and 2 bleaching steps), and more steps led to larger loss of material. MAE and CHE are distinguished by the heating method: MAE employs an indirect heating via microwaves, while CHE relies on direct heating through irradiation within the infrared region. In both cases, the use of closed vessels (autoclaves) leads to an increase of the extraction process efficacy, enabling more direct isolation of the desired product. From a practical standpoint, CHE involving classical heating appears more feasible instrumentally while also ensuring higher efficiency in biopolymer isolation.

3.3. Cellulose characterization

3.3.1. Cellulose composition, structure and surface morphology

The chemical structure of the extracted cellulose samples was determined by CP-MAS ^{13}C NMR spectroscopy (Fig. 4). The various carbon atoms on the β-d-glucose ring of cellulose are identified: signals at 64, 72–74, 82–88, and 105 ppm belong to C_6, $C_{2,3,5}$, C_4 and C_1, respectively. These findings are in agreement with literature data [50,51], suggesting that the structure of cellulose was preserved. In other words, the treatments performed in the various extractions did not damage the chain structure of cellulose. However, other signals are observed at 152, 147, 135 and 55 ppm, which are assigned to the aromatic groups of lignin although the content of lignin could not be quantified. More bleaching steps could allow to remove residual lignin, but this may deteriorate the structure of the cellulose, not to mention increased environmental and energy costs.

ATR-FTIR was also used to investigate the chemical composition of the extracted cellulose samples, as shown in Fig. 5. The spectra of the various polysaccharides present similar profiles with little difference in the intensity of peaks. Comparison with the starting Mix_OSF reveals the disappearance of the peak at 1750 cm^{-1} and the intensity decrease of the peak 1230 cm^{-1} belonging to the acetyl groups of hemicellulose and lignin [52], respectively, confirming the total or partial removal of these components. The characteristic peaks of cellulose were observed, including a large hydroxyl stretching band at 3320 cm^{-1}, a C-H stretching vibration at ~1420 cm^{-1} associated with cellulose crystalline domains [53], an intense peak at ~1030 cm^{-1} due to C-O-C bonds, and a C-H bending peak at ~894 cm^{-1} involving β-glycosidic bonds of cellulose. The peak at 1590 cm^{-1} is characteristic of lignin -C═O bonds [6,52,54], indicating the presence of residual lignin in the extracted cellulose, which is in agreement with CP-MAS ^{13}C NMR results (Fig. 4). The intensity of the lignin peak at 1590 cm^{-1} is lower in the CHE-extracted samples than in the HWE-extracted ones which have a broader and more intense peak (Fig. 5), showing a notable difference in the residual lignin content according to the extraction methods used. These findings suggest that CHE is the most suitable method, allowing better removal of lignin and thus producing cellulose of higher purity. In addition, FT-IR confirmed the chemical structure of extracted cellulose, and the effectiveness of the treatments in removing hemicellulose and phenolic compounds.

The surface morphology of the different cellulose samples was examined by SEM, as shown in Fig. 6. Particles of tens of micrometers

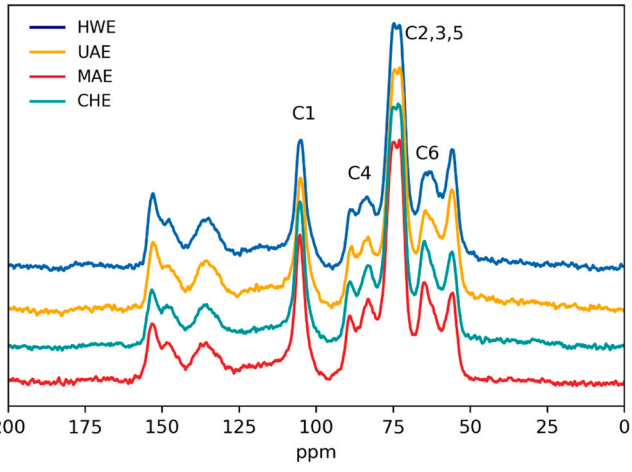


Fig. 4. CP-MAS ^{13}C NMR spectra of the cellulose samples obtained by using different extraction techniques.

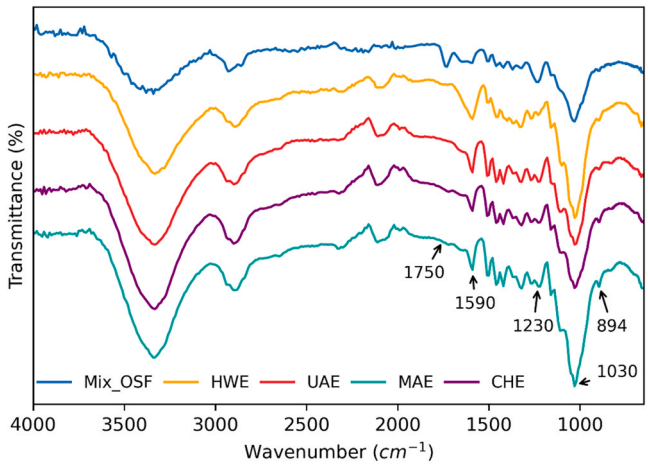


Fig. 5. FTIR spectra of the different samples of extracted celluloses compared with the original olive stone flour (Mix_OSF).

are observed in all cases, showing that the extraction techniques have little impact on the morphology of extracted cellulose. Furthermore, the samples are morphologically comparable with those obtained in the work by Garnett et al. [6], and with commercial microcrystalline cellulose, reinforcing the effectiveness of the extraction techniques.

3.3.2. *Thermal stability*

The thermal stability of the extracted celluloses were determined using TGA as shown in Figure A.2 and the results of measurements are summarized in Table 1.

The onset decomposition temperature varies according to the extraction technique used, ranging from 274°C for HWE to 316°C for CHE, showing that cellulose extracted by conventional heating presents better resistance to thermal decomposition compared with other extraction methods. This can be explained by the fact that the celluloses extracted by MAE and CHE were subjected to less treatment steps compared to those obtained by HWE and UAE, which could preserve the cellulose structure. A notable difference in the remaining mass values is also noticed between the samples, ranging from 20 % for HWE to 5 % for CHE at 800°C. This difference could be explained by the production of biochars during decomposition of cellulose or lignin between 450 and 600°C [55,56]. Previous studies have shown that, compared to cellulose,

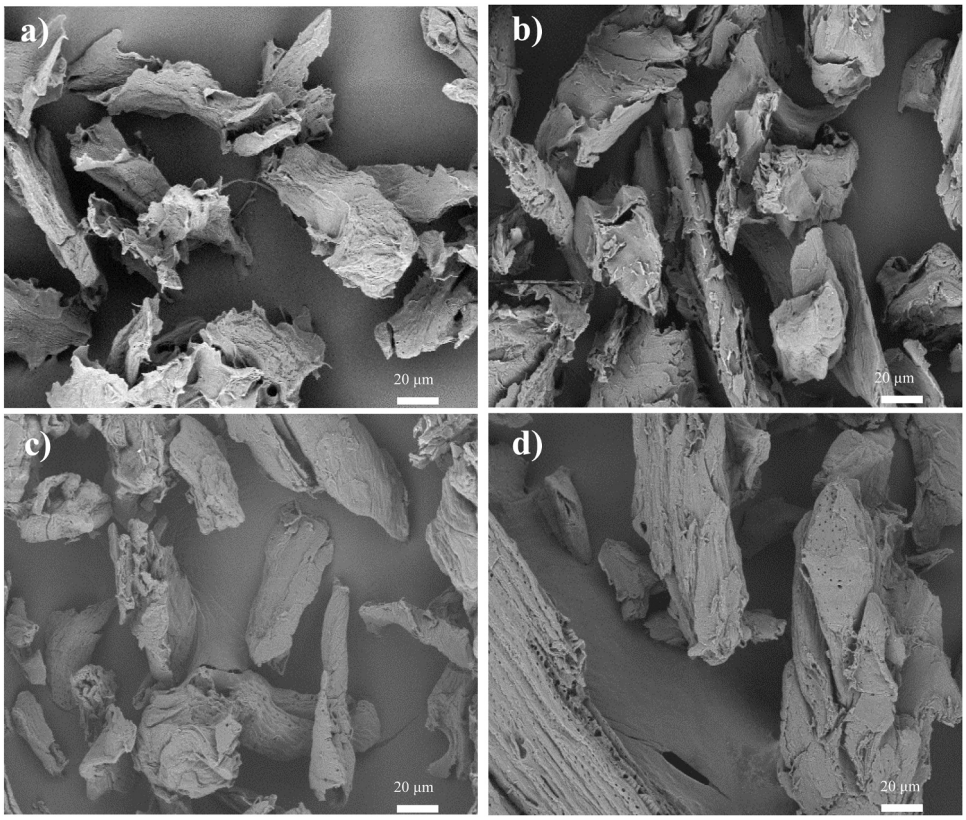


Fig. 6. SEM images of celluloses samples: a) HWE, b) UAE, c) MAE and d) CHE.

Table 1
Thermal stability of celluloses extracted by using different methods, i.e. hot water extraction (HWE), ultrasound-assisted extraction (UAE), microwave-assisted extraction (MAE) and conventional heating extraction (CHE).

Extraction method	Onset decomposition temperature (°C)	Maximum loss temperature (°C)	Mass loss (%)	Remaining mass (%) at		
				450°C	600°C	800°C
HWE	274	324	64	29	26	20
UAE	288	332	71	25	21	13
MAE	295	334	77	20	17	12
CHE	316	361	77	19	15	5

lignin decomposition led to higher biochar yields due to its thermally stable aromatic ring structure [57,58]. The lower remaining mass in MAE and CHE extracted celluloses suggest a lower lignin content than in those obtained by HWE and UAE, supporting the hypothesis that MAE and CHE promote more efficient lignin removal. Additionally, this difference may indicate the selective extraction of distinct lignin fractions with varying thermal stability [59]. Thus, the remaining mass in TGA can serve as an indicator of lignin content, reinforcing the effectiveness of MAE and CHE in cellulose purification, in agreement with the results obtained from the FTIR spectra (Section 3.3.1).

3.3.3. X-ray diffraction (XRD)

The crystalline structure of the extracted celluloses was determined by XRD, as shown in Fig. 7. Similar XRD patterns are observed for all samples, including a large halo from amorphous domains and three major diffraction peaks at 17.0, 23.0 and 36.5° from crystalline domains. Nevertheless, a significant difference is detected between the peak intensities of the four samples, indicating a more ordered structure in cellulose samples from MAE and CHE as compared to that of HWE and UAE samples. The crystallinity index (CI) of each cellulose sample was determined according to the method Section 2.3.6, [34,35], and the obtained data are summarized in Table A.6. A noticeable difference of CI is observed between the four cellulose samples. The CI is higher for MAE and CHE (44 and 45 % respectively) than for HWE and UAE (37 and 40 %, respectively). These values are lower than those reported in literature [34,35], probably due to the presence of non-cellulosic compounds such as lignin which decreases the overall crystallinity as reported by Ravindran et al. Ravindran et al. [35]. These findings well corroborate with the TGA data shown above. In fact, CHE and MAE exhibit a higher onset composition temperature (see Table 1) due to their more pronounced crystalline character, making them more resistant to thermal degradation. This further supports the hypothesis that MAE and CHE promote more efficient lignin removal compare to HWE and UAE, resulting in a more crystalline and thermally stable cellulose structure.

4. Conclusion

In this work, different varieties of olive stones were used to evaluate their potential as a raw material to produce value added compounds. Comparing the composition of Chemlalie and Koroneiki olive stones with a mixture of varieties (Mix_OSF) showed that using the latter was highly relevant for cellulose extraction as it had the highest dietary fiber content (86 %). From the angle of circular economy, using a mixture simplifies the recovery of olive stones for valorization.

Extraction of cellulose was performed using four different techniques, including hot water extraction (HWE), ultrasound-assisted extraction (UAE), microwave-assisted extraction (MAE) and conventional heating extraction (CHE) in order to explore an innovative approach for obtaining cellulose with enhanced physicochemical characteristics suitable for advanced material applications. MAE and CHE methods led to significantly higher yields than HWE and UAE ones, while reducing the reaction time and the amount of solvent used. Structural analyses showed that cellulose could be isolated while preserving its structure after the various treatments, including acid treatment, alkaline treatment and bleaching. NMR and FT-IR revealed the presence of residual lignin in extracted cellulose samples. MAE and CHE methods also yielded cellulose with higher crystallinity index and better thermal stability due to lower lignin contents.

Therefore, MAE and CHE emerge as promising and efficient techniques for cellulose extraction from olive stones. This abundant and cost-effective waste has significant potential for valorization and can play a role in a circular economy, offering prospects for large-scale implementation and exploring possible applications for extracted cellulose in the healthcare and pharmaceutical sectors.

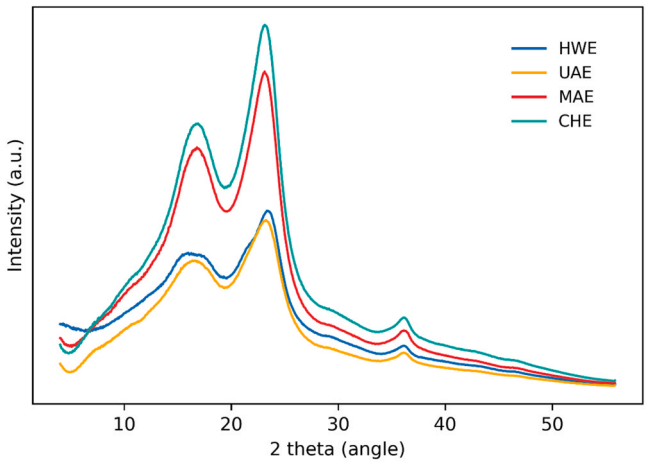


Fig. 7. XRD patterns of celluloses extracted by hot water extraction (HWE), ultrasound-assisted extraction (UAE), microwave-assisted extraction (MAE) and conventional heating extraction (CHE).

CRediT authorship contribution statement

Mohamed Chamkha: Resources. **Amina Maalej:** Resources. **Céline Pochat-Bohatier:** Writing – review & editing, Supervision. **Martin Drobek:** Resources, Data curation. **Hana Maalej:** Writing – review & editing, Project administration. **Anne-Constance Macarez:** Writing – original draft, Methodology, Investigation, Data curation. **Suming Li:** Writing – review & editing, Supervision, Funding acquisition.

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.jece.2025.118204.

Data Availability

Data will be made available on request.

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